16 Normal Modes pt 4 Diff Eqs

Some notes ahead of time

Pre notes

From differential equations, the reader should know the following:

Solving Homogenous ode:

$$\ddot{x}_n + 2\beta\dot{x}_n + \omega_0^2 x_n = 0$$

Finding a particular solution x_p

$$\ddot{x}_p + 2\beta\dot{x}_p + \omega_0^2 x_p = \frac{F_{\text{drive}}(t)}{m}$$

The general solution is

$$x(t) = x_n(t) + x_p(t)$$

Homogenous solution

We have underdamped, overdamped, and critically damped systems.

$$x(t) = \begin{cases} e^{-\beta t} C \cos[\sqrt{\omega_0^2 - \beta^2} t + \delta], & \beta < \omega_0 \text{ Underdamped} \\ Ae^{-(\beta+\sqrt{\beta^2-\omega_0^2})t} + Be^{-(\beta-\sqrt{\beta^2\omega_0^2})t}, & \beta > \omega_0 \text{ Overdamped} \\ Ae^{-\beta t} + Bte^{-bt}, & \beta = \omega_0 \text{ Critically Damped} \end{cases}$$

All of these damp to zero for $\beta t \gg 1$

Damped Driven Oscillators

Lets let there be some amount of friction

$$\vec{F}_{\text{friction}} = -b\vec{v}$$

We can't do Lagrangian mechanics. This is painful. Lets use regular Newtonian mechanics. Draw a free body diagram with friction opposing motion and a net spring constant back towards equilibrium.

Lets add some external driving to make it fun: Lets drive the left wall as a function of time with $\vec{F}_{\text{drive}}(t)$.
The force from the driven wall will transmute through the springs.

$$m\ddot{x} = -b\dot{x} - kx - k'x + F_{\text{drive}}$$

We can rewrite this

$$\ddot{x} = -\underbrace{\frac{b}{m}}_{2\beta}\dot{x} - \underbrace{\frac{k+k'}{m}}_{\omega_0^2}x + \frac{F_{\text{drive}}}{m}$$

β and ω_0 have dimensions of frequency, $\frac{1}{\text{time}}, s^{-1}$.

For now, lets take a very special $F_{\text{drive}}(t)$ to make it easy to solve, and then we can make it general.
Let $F_{\text{drive}}(t) = f_0 \cos(\omega t)$.

The system is

$$\ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f_0 \cos(\omega t)$$

Lets make this COMPLEX.

$$\ddot{z} + 2\beta\dot{z} + \omega_0^2 z = f_0 e^{i\omega t}$$

Where our solution is just $\text{Re}[z(t)]$

We make the Ansatz $z(t) = Ae^{i\omega t} = |A|e^{i(\omega t + \delta)}$, where $A = |A|e^{i\delta}$.

Lets use this solution in the differential equation.

$$-\omega^2 Ae^{i\omega t} + 2\beta(i\omega)Ae^{i\omega t} + \omega_0^2 Ae^{i\omega t} = f_0 e^{i\omega t}$$

We get now

$$A(\omega_0^2 - \omega^2 + 2i\beta\omega) = f_0$$
$$A = \frac{f_0}{(\omega_0^2 - \omega^2) + 2i\beta\omega}$$

This has a magnitude $|A|$ and phase δ. This is just some complex number. Lets take that solution to get the amplitude and phase information.
How do we solve for δ?
We can write A as real and imaginary.

$$\text{let } A = a + ib$$

$$a = \text{Re}(A), b = \text{Im}(A)$$

$$\tan\delta = \frac{\text{Im}(A)}{\text{Re}(A)}$$

We can simplify by multiplying A by $\frac{[(\omega_0^2-\omega^2)-2i\beta\omega]}{[(\omega_0^2-\omega^2)-2i\beta\omega]}$. This is just 1, but will give us a nice real and imaginary part.

$$A = \frac{f_0(\omega_0^2 - \omega^2)}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2} + i\frac{f_0(-2\beta\omega f_0)}{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}$$

$$|A| = \sqrt{a^2 + b^2}$$

Lets take these monstrosities.

$$\sqrt{\frac{f_0^2(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2 f_0^2}{[(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2]^2}}$$

This simplifies nicely

$$|A| = \frac{f_0^2}{\sqrt{(\omega_0^2 - \omega^2)^2 + 4\beta^2\omega^2}}$$

We also have

$$\begin{aligned}\tan\delta &= \frac{\mathrm{Im}(A)}{\mathrm{Re}(A)} \\ &= \frac{(-2\beta\omega f_0)}{f_0(\omega_0^2-\omega^2)}\end{aligned}$$

$$= \frac{2\beta\omega}{(\omega_0^2-\omega^2)^2}$$

We find the quadrant from here

Note that this has no free parameters.

$$z(t)_p = |A|e^{i(\omega t+\delta)}$$

Where the amplitude and phase only depend on system parameters.

This is the particular solution that the system converges to over time, we will decay to exactly this.

Amplitude:

At $\omega = \omega_0$, we'll have $A = \frac{f_0}{2\beta\omega}$. For large mismatch between ω and ω_0, we'll have $A = \frac{f_0}{\sqrt{\omega^4+4\beta^2\omega^2}}$

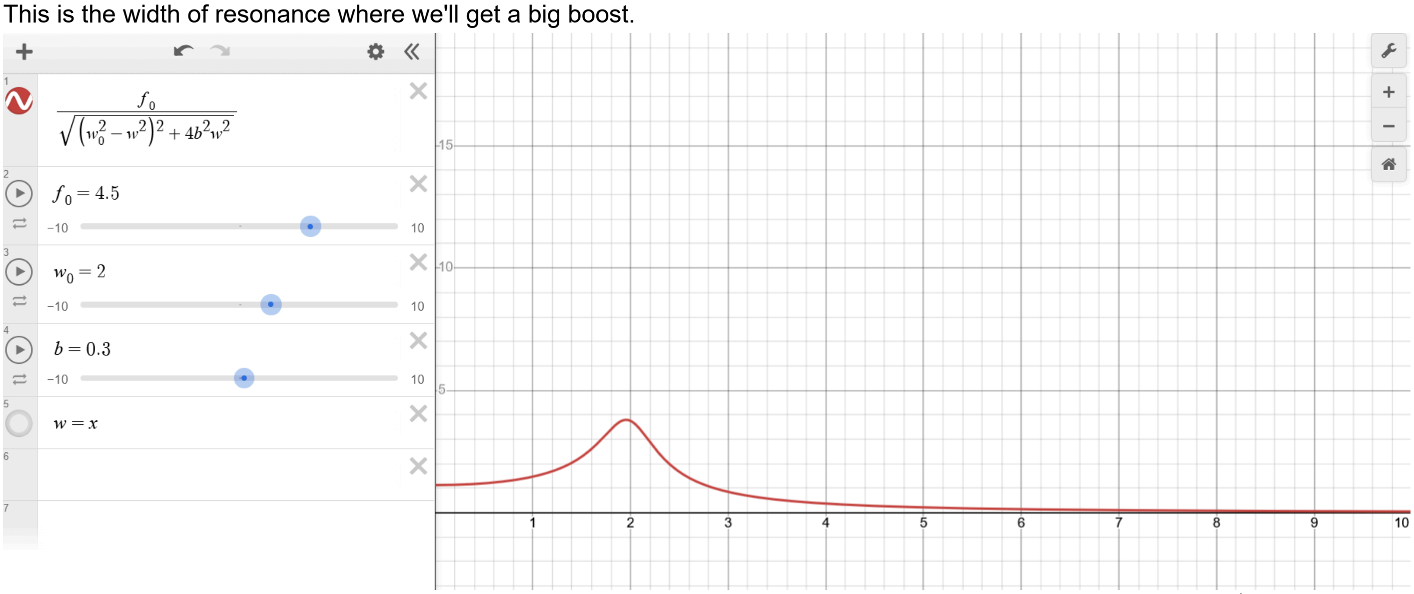
For ω_0 to be in resonance, we need

$$\begin{aligned}(\omega_0^2-\omega^2)^2 &\leq 4\beta^2\omega^2 \\ \left|\omega_0^2-\omega^2\right| &\leq 2\beta\omega \\ (\omega_0+\omega)\,|\omega_0-\omega| &\leq 2\beta\omega\end{aligned}$$

Because we want $\omega_0 \approx \omega$, this becomes

$$|\omega_0-\omega| \leq \beta$$

This is the width of resonance where we'll get a big boost.



As β goes to zero, we have no damping and infinite response. We no longer have the approximation of $U_{\text{sho}} = \frac{1}{2}kx^2$, and get nonlinear dynamics. Friction makes life nicer.

General Driving

We can represent any periodic

$$F_{\text{drive}}(t) = \mathrm{Re}\left[\sum_j A_j e^{i\omega_j t}\right]$$

Only one term will have $\omega_j = \omega_0$: Where driving will match the natural frequency. We can generally ignore any frequency components that don't match the natural frequency.